Project 4:Fake News Prediction

About the Dataset:

1.id:unique id for a news article 2.title:the title of a news artical 3.author:author of the news artical 4.text:the text of the artical;could be incomplete 5.label:a label that marks whether the newsarticle is real or fake

1:Fake news 0:real News

importing the Dependencies

In [1]:
```
import numpy as np
import pandas as pd
import re
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

In [2]:
```
import nltk
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[2]: True

In [3]:
```
print(stopwords.words('english'))
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'a
nd', 'any', 'are', 'aren', "aren't", 'as', 'at', 'be', 'because', 'been', 'befor
e', 'being', 'below', 'between', 'both', 'but', 'by', 'can', 'couldn', "could
n't", 'd', 'did', 'didn', "didn't", 'do', 'does', 'doesn', "doesn't", 'doing', 'd
on', "don't", 'down', 'during', 'each', 'few', 'for', 'from', 'further', 'had',
'hadn', "hadn't", 'has', 'hasn', "hasn't", 'have', 'haven', "haven't", 'having',
'he', "he'd", "he'll", 'her', 'here', 'hers', 'herself', "he's", 'him', 'himsel
f', 'his', 'how', 'i', "i'd", 'if', "i'll", "i'm", 'in', 'into', 'is', 'isn', "is
n't", 'it', "it'd", "it'll", "it's", 'its', 'itself', "i've", 'just', 'll', 'm',
'ma', 'me', 'mightn', "mightn't", 'more', 'most', 'mustn', "mustn't", 'my', 'myse
lf', 'needn', "needn't", 'no', 'nor', 'not', 'now', 'o', 'of', 'off', 'on', 'onc
e', 'only', 'or', 'other', 'our', 'ours', 'ourselves', 'out', 'over', 'own', 'r
e', 's', 'same', 'shan', "shan't", 'she', "she'd", "she'll", "she's", 'should',
'shouldn', "shouldn't", "should've", 'so', 'some', 'such', 't', 'than', 'that',
"that'll", 'the', 'their', 'theirs', 'them', 'themselves', 'then', 'there', 'thes
e', 'they', "they'd", "they'll", "they're", "they've", 'this', 'those', 'throug
h', 'to', 'too', 'under', 'until', 'up', 've', 'very', 'was', 'wasn', "wasn't",
'we', "we'd", "we'll", "we're", 'were', 'weren', "weren't", "we've", 'what', 'whe
n', 'where', 'which', 'while', 'who', 'whom', 'why', 'will', 'with', 'won', "wo
n't", 'wouldn', "wouldn't", 'y', 'you', "you'd", "you'll", 'your', "you're", 'you
rs', 'yourself', 'yourselves', "you've"]
```

Data Pre-processing

In [4]:
```
#loading the dataset to a pandas DataFrame
news_dataset=pd.read_csv(r'D:\Projects\archive (3)\News _dataset\Fake.csv')
```

In [5]:
```
news_dataset.shape
```

Out[5]:
```
(23481, 4)
```

In [6]:
```
news_dataset.head()
```

Out[6]:

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

In [7]:
```
news_dataset.isnull().sum()
```

Out[7]:
```
title      0
text       0
subject    0
date       0
dtype: int64
```

In [8]:
```
news_dataset=news_dataset.fillna('')
```

In [9]:
```
news_dataset['content']=news_dataset['title']
```

In [10]:
```
print(news_dataset['content'])
```

```
0         Donald Trump Sends Out Embarrassing New Year’...
1         Drunk Bragging Trump Staffer Started Russian ...
2         Sheriff David Clarke Becomes An Internet Joke...
3         Trump Is So Obsessed He Even Has Obama’s Name...
4         Pope Francis Just Called Out Donald Trump Dur...
                               ...
23476    McPain: John McCain Furious That Iran Treated ...
23477    JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
23478    Sunnistan: US and Allied ‘Safe Zone’ Plan to T...
23479    How to Blow $700 Million: Al Jazeera America F...
23480    10 U.S. Navy Sailors Held by Iranian Military ...
Name: content, Length: 23481, dtype: object
```

In [11]:
```
x=news_dataset.drop(columns='subject',axis=1)
y=news_dataset['subject']
print(x)
print(y)
```

```
                                                   title  \
0       Donald Trump Sends Out Embarrassing New Year’...
1       Drunk Bragging Trump Staffer Started Russian ...
2       Sheriff David Clarke Becomes An Internet Joke...
3       Trump Is So Obsessed He Even Has Obama’s Name...
4       Pope Francis Just Called Out Donald Trump Dur...
...                                                  ...
23476  McPain: John McCain Furious That Iran Treated ...
23477  JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
23478  Sunnistan: US and Allied ‘Safe Zone’ Plan to T...
23479  How to Blow $700 Million: Al Jazeera America F...
23480  10 U.S. Navy Sailors Held by Iranian Military ...

                                                    text               date  \
0      Donald Trump just couldn t wish all Americans ...  December 31, 2017
1      House Intelligence Committee Chairman Devin Nu...  December 31, 2017
2      On Friday, it was revealed that former Milwauk...  December 30, 2017
3      On Christmas day, Donald Trump announced that ...  December 29, 2017
4      Pope Francis used his annual Christmas Day mes...  December 25, 2017
...                                                  ...                ...
23476  21st Century Wire says As 21WIRE reported earl...   January 16, 2016
23477  21st Century Wire says It s a familiar theme. ...   January 16, 2016
23478  Patrick Henningsen  21st Century WireRemember ...   January 15, 2016
23479  21st Century Wire says Al Jazeera America will...   January 14, 2016
23480  21st Century Wire says As 21WIRE predicted in ...   January 12, 2016

                                                 content
0       Donald Trump Sends Out Embarrassing New Year’...
1       Drunk Bragging Trump Staffer Started Russian ...
2       Sheriff David Clarke Becomes An Internet Joke...
3       Trump Is So Obsessed He Even Has Obama’s Name...
4       Pope Francis Just Called Out Donald Trump Dur...
...                                                  ...
23476  McPain: John McCain Furious That Iran Treated ...
23477  JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
23478  Sunnistan: US and Allied ‘Safe Zone’ Plan to T...
23479  How to Blow $700 Million: Al Jazeera America F...
23480  10 U.S. Navy Sailors Held by Iranian Military ...

[23481 rows x 4 columns]
0               News
1               News
2               News
3               News
4               News
            ...
23476    Middle-east
23477    Middle-east
23478    Middle-east
23479    Middle-east
23480    Middle-east
Name: subject, Length: 23481, dtype: object
```

Stemming:

Stemming is the process of reducing a word to its Root word

example:actor,actress,acting->act

In [12]:
```
port_stem=PorterStemmer()

def stemming(content):
    stemmed_content=re.sub('[^a-zA-z]','',content)
    stemmed_content=stemmed_content.lower()
    stemmed_content=stemmed_content.split()
    stemmed_content=[port_stem.stem(word) for word in stemmed_content if not wor
    stemmed_content=''.join(stemmed_content)
    return stemmed_content
```

In [13]:
```
news_dataset['content']=news_dataset['content'].apply(stemming)
```

In [14]:
```
print(news_dataset['content'])
```

```
0        donaldtrumpsendsoutembarrassingnewyearsevemess...
1        drunkbraggingtrumpstafferstartedrussiancollusi...
2        sheriffdavidclarkebecomesaninternetjokeforthre...
3        trumpissoobsessedheevenhasobamasnamecodedintoh...
4        popefrancisjustcalledoutdonaldtrumpduringhisch...
                               ...
23476    mcpainjohnmccainfuriousthatirantreatedussailor...
23477    justiceyahoosettlesemailprivacyclassactionmfor...
23478    sunnistanusandalliedsafezoneplantotaketerritor...
23479    howtoblowmillionaljazeeraamericafinallycallsit...
23480    usnavysailorsheldbyiranianmilitarysignsofaneoc...
Name: content, Length: 23481, dtype: object
```

In [15]:
```
#separating the data and label
x=news_dataset['content'].values
y=news_dataset['subject']
print(x)
print(y)
```

```
['donaldtrumpsendsoutembarrassingnewyearsevemessagethisisdisturb'
 'drunkbraggingtrumpstafferstartedrussiancollusioninvestig'
 'sheriffdavidclarkebecomesaninternetjokeforthreateningtopokepeopleintheey'
 ...
 'sunnistanusandalliedsafezoneplantotaketerritorialbootyinnorthernsyria'
 'howtoblowmillionaljazeeraamericafinallycallsitquit'
 'usnavysailorsheldbyiranianmilitarysignsofaneoconpoliticalstunt']
0               News
1               News
2               News
3               News
4               News
            ...
23476    Middle-east
23477    Middle-east
23478    Middle-east
23479    Middle-east
23480    Middle-east
Name: subject, Length: 23481, dtype: object
```

In [16]:
```
#separating the data and subject
x=news_dataset['content'].values
y=news_dataset['subject'].values
```

In [ ]:
```
#converting the textual data to numerical data
vectorizer=TfidfVectorizer()
```

```
x=vectorizer.fit_transform(x)
```

In [17]:
```
print(x)
```

```
['donaldtrumpsendsoutembarrassingnewyearsevemessagethisisdisturb'
 'drunkbraggingtrumpstafferstartedrussiancollusioninvestig'
 'sheriffdavidclarkebecomesaninternetjokeforthreateningtopokepeopleintheey'
 ...
 'sunnistanusandalliedsafezoneplantotaketerritorialbootyinnorthernsyria'
 'howtoblowmillionaljazeeraamericafinallycallsitquit'
 'usnavysailorsheldbyiranianmilitarysignsofaneoconpoliticalstunt']
```

In [18]:
```
print(y)
```

```
['News' 'News' 'News' ... 'Middle-east' 'Middle-east' 'Middle-east']
```

In [19]:
```
y.shape
```

Out[19]:
```
(23481,)
```

In [23]:
```
print(x)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
	with 28854 stored elements and shape (23481, 18016)>
  Coords	Values
  (0, 3947)	1.0
  (1, 4056)	1.0
  (2, 11614)	1.0
  (3, 13892)	1.0
  (4, 10151)	1.0
  (5, 10491)	1.0
  (6, 5092)	1.0
  (7, 14223)	1.0
  (8, 4841)	1.0
  (9, 15728)	1.0
  (10, 9887)	1.0
  (11, 16408)	1.0
  (12, 753)	1.0
  (13, 16246)	1.0
  (14, 5673)	1.0
  (15, 13197)	1.0
  (16, 12548)	1.0
  (17, 8609)	1.0
  (18, 11792)	1.0
  (19, 10878)	1.0
  (20, 6546)	1.0
  (21, 7407)	1.0
  (22, 8203)	1.0
  (23, 2811)	1.0
  (24, 17155)	1.0
  :	:
  (23456, 4418)	1.0
  (23457, 43)	1.0
  (23458, 4237)	1.0
  (23459, 4555)	1.0
  (23460, 13311)	1.0
  (23461, 10717)	1.0
  (23462, 1335)	1.0
  (23463, 4419)	1.0
  (23464, 4251)	1.0
  (23465, 12694)	1.0
  (23466, 1521)	1.0
  (23467, 9784)	1.0
  (23468, 11114)	1.0
  (23469, 1263)	1.0
  (23470, 573)	1.0
  (23471, 11570)	1.0
  (23472, 5608)	1.0
  (23473, 631)	1.0
  (23474, 12630)	1.0
  (23475, 5918)	1.0
  (23476, 8136)	1.0
  (23477, 7166)	1.0
  (23478, 12087)	1.0
  (23479, 6340)	1.0
  (23480, 15203)	1.0
```

Splitting the dataset to training & test data

In [24]:
```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,stratify=y,rand
```

Training the Model:Logistic Regression

In [26]:
```
model=LogisticRegression()
```

In [27]:
```
model.fit(x_train,y_train)
```

Out[27]:

▾ LogisticRegression ⓘ ?

▸ Parameters

▸ Fitted attributes

Evaluation

accuracy score

In [28]:
```
#accuracy score on the training data
x_train_prediction=model.predict(x_train)
training_data_accuracy=accuracy_score(x_train_prediction,y_train)
```

In [29]:
```
print('Accuracy score of the training data:',training_data_accuracy)
```

```
Accuracy score of the training data: 0.6993718057921635
```

In [34]:
```
#accuracy score on the test data
x_test_prediction=model.predict(x_test)
test_data_accuracy=accuracy_score(x_test_prediction,y_test)
```

In [35]:
```
print('Accuracy score of the test data:',test_data_accuracy)
```

```
Accuracy score of the test data: 0.46604215456674475
```

Making a Predictive System

In [36]:
```
x_new=x_test[3]

prediction=model.predict(x_new)
print(prediction)

if(prediction[0]==[0]):
    print('The news is Real')
else:
    print('the news is Fake')
```

```
['politics']
the news is Fake
```

In [41]:
```
print(y_test[4])
```

```
politics
```